

Protocol 1

Pharmacophore models

General Instructions,

1. Each student is required to complete their assigned task, and all tasks within the group should be compiled and submitted as a single assignment per group.
2. Students are encouraged to follow the provided protocol. The instructor will spend 10 minutes with each group to discuss the developed models and their outcomes.
3. The submission deadline is **18th October 2024**.
4. Each group should create a folder for the assigned tasks and submit a soft copy of the folder to my TA (further details will be shared shortly).

Data Provided:

- Perform Training Using the Template
- Use the remaining data for validation
- Use the Unknown data for prediction

Data Description:

Provided is the dataset of the inhibitors of hERG. With potency ranging from 0.0267-82.688 Nm. You are required to generate the Pharmacophore hypothesis of hERG inhibition using provided data. You are also required to predict the hERG liability of the data given as prediction set.

You are guided to follow the following steps:

Step 1: Building of a conformation-database: (5.0)

MOE/Compute/Conformations/Conformation Import...

Use default values (exception: deactivate Split Output, deactivate Input Filter with Clear), add the mdb file as input file.

Pack the database (all conformations in one line): type the following command in the MOE commandline: `db_ConfPack ['conf_out.mdb', 'dbname_conf.mdb']`

To get the activity values merge the new conformational database with the original one:

Go to DBV/File/Merge... in the packed database.

Choose the original database as database 2, deactivate "New Database".

Step 2: Building of a pharmacophore model:

In the original database: look at the structures in the database browser (DBV/File/Browser...). Choose an active compound (Template) for the building of the model and copy it into the MOE window.

Hide the hydrogen atoms: MOE/Render/Hide/Hydrogens

Open the Pharmacophore Query Editor: MOE/File/New/Pharmacophore Query...

Change the Ligand Annotation to "Long" (for showing the labels).

An explanation for the abbreviations can be found under Info...

(a) Build a pharmacophore query: (10.0)

Select a ligand annotation point and click on Create: Feature in the Query Editor.

The radius can be changed with the wheel next to "R:".

The appearance can be changed with Change... under Query Rendering.

Query search:

Click on Search...

Choose your packed conformational database.

If the database is not opened click on Open... in the upper right corner of the window.

Choose:

Hit Entries: Select

Results: No Output

A click on Search starts the query search. You should find as many as possible from the active compounds, and as less as possible from the inactives.

Possible changes in the pharmacophore query:

(b) Not enough compounds were selected:

- increase the radius of the features
- decrease the number of features (by activating Ignore), maybe try first with Enable Partial Match if the search works better with less features

(c) Too many compounds were selected:

- decrease the radius of the features
- choose additional features

(d) Output of the identified conformations:

In the Pharmacophore Search window choose:

- Results: Conformations
- Setup>> Code and EC50

(e) Choose the best conformations:

Sort the conformations in ph4out.mdb with DBV/Compute/Sort

(Priority 1: mseq; Priority 2: rmsd)

Choose the best entries again in DBV/Compute/Sort:

deactivate Sort Entries; activate Select Unique Entries; choose Code or mseq

Hide the additional entries with DBV/Entry/Hide Unselected Entries.

Use the Database Browser to view the best conformations in the MOE window.

Step 3: Model Comparison: (10.0)

Sr #	Model Name	Model Statistics			Features Selected	Distance Matrix				
		Accuracy	Specificity	Sensitivity			Acc	Don	Hyd	Aro
1	Ali				Acc, don, hyd, Aro		Acc	Don	Hyd	Aro
						Acc	0			
						don		0		
						hyd			0	
2	Ahmad					aro				0
3	Saba									
4	Ayesha									
5	Fatima									

Step 4: Predict the hETRG Liability of the given unknown data. (5)

Protocol: (5x2= 10.0)

Discussion of the all the developed models and advocate the finally selected model, in your discuss focus on the following points for each model,

- Which compound/compounds were used to build the query?
- Which features were used? In which distance to each other (use MOE/Edit/Measure/ Distances...) are the features? Include a picture.
- Can the model distinguish between active and inactive compounds? Which compounds were identified, which not?
- Are similar compounds identified in a similar manner? Include some pictures of the identified conformations.
- Explain the predicted behaviour of the compounds in the unknown data.

Send the **final pharmacophore query file** (.ph4): click Save in the Pharmacophore Query Editor.